



Melbourne Housing Price Prediction

Machine Learning Project Report

Supervised by:

Eng. Nagham Yehya

Prepared by:

Ziad Reda Mohamed Fahmy – 231014601

AbdulRahman Mohamed Farag – 231015031

AbdulRahman Mohamed Hussein - 231015232

Academic Year 2025 – 2026

1. Introduction

The real estate market is one of the most economically significant sectors globally. Accurate property price prediction is valuable to buyers, sellers, investors, and financial institutions. Melbourne, Australia, has one of the most dynamic housing markets in the Asia-Pacific region, making it an ideal subject for a predictive modelling study.

This report documents the complete machine learning pipeline developed to predict Melbourne residential house prices. The workflow covers exploratory data analysis, data cleaning and preprocessing, model training, comparison of four regression algorithms, and a final evaluation of results.

2. Problem Statement

Predicting real estate prices is a complex regression problem influenced by a wide range of factors: structural attributes of the property (number of rooms, building area), location (suburb, distance to CBD, council area), and market conditions (method of sale, seller agency).

The goal of this project is to build a regression model that can accurately estimate the sale price of a Melbourne residential property given a set of known attributes, while handling real-world data quality issues such as missing values, outliers, and categorical variables.

3. Dataset Description

The Melbourne Housing dataset is a publicly available collection of residential property sales records scraped from Domain.com.au. It contains approximately 13,580 records with 21 features covering property characteristics, location, and sale information. The target variable is Price (sale price in AUD).

Feature	Type	Description
Suburb	Categorical	Name of the suburb
Rooms	Numeric	Number of rooms
Type	Categorical	h = house, t = townhouse, u = unit
Price	Numeric	Sale price in AUD (target variable)
Method	Categorical	Method of sale
Distance	Numeric	Distance from Melbourne CBD (km)
Bedroom2	Numeric	Number of bedrooms
Bathroom	Numeric	Number of bathrooms
Car	Numeric	Number of car spaces
Landsize	Numeric	Land size in square metres
BuildingArea	Numeric	Building size in square metres
YearBuilt	Numeric	Year the property was built
CouncilArea	Categorical	Governing council for the area

Regionname	Categorical	General region of Melbourne
------------	-------------	-----------------------------

The dataset had significant missing values: BuildingArea (~47%), YearBuilt (~39%), CouncilArea (~5%), and Car (~1%).

4. Data Exploration

Exploratory data analysis (EDA) was conducted to understand the distribution, relationships, and quality of the dataset before preprocessing.

4.1 Target Variable

The Price column is heavily right-skewed. Most properties sold between \$400,000 and \$1,500,000 AUD, but a long tail of luxury properties extends to over \$9,000,000. This skew confirmed the need for outlier treatment.

4.2 Correlation Analysis

A correlation heatmap showed that Rooms, Bedroom2, Bathroom, and Car are positively correlated with Price. Distance from the CBD shows a moderate negative correlation — properties closer to the city command higher prices.

4.3 Property Types

The dataset is dominated by houses (h) at approximately 70%, followed by units (u) at 22%, and townhouses (t) at 8%.

5. Data Visualization

The following visualizations were generated to support data understanding:

- Correlation Heatmap: showed Rooms, Bathroom, and BuildingArea as the strongest numerical predictors of Price.
- Price Histogram: confirmed a right-skewed distribution with most sales concentrated below \$1.5M.
- Boxplot of Price by Rooms (log scale): showed a clear positive relationship between room count and price, with increasing variance for larger properties.
- Pie Chart of Property Type: visualized the dataset composition (houses 70%, units 22%, townhouses 8%).
- Scatter plot of Price vs. Building Area (log-log scale): showed a positive trend with high variance, indicating that size alone does not determine price.
- Bar chart of Top 15 Suburbs: identified the most represented suburbs including Reservoir, Bentleigh East, and Richmond.
- Boxplots of all numeric features: revealed outliers across Price, Landsize, BuildingArea, and other columns.

6. Data Preprocessing

6.1 Missing Value Imputation

Context-aware, group-based imputation was applied rather than global statistics, to preserve the natural structure of the data:

- Car: filled with the global mode, as car space count is relatively uniform across the dataset.
- BuildingArea: filled with the median per Rooms group, since larger homes naturally have more rooms. Values exceeding Landsize were then capped at Landsize to remove physically impossible entries.
- YearBuilt: filled with the median per Suburb, since properties in the same area tend to be built in similar eras.
- CouncilArea: filled with the mode per Suburb, as properties in the same suburb almost always fall under the same council.

6.2 Outlier Detection and Correction

The IQR (Interquartile Range) method was applied to all numerical features. Rather than removing outlier records, values were clipped to the fence boundaries ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$). This preserved all records while reducing the distorting effect of extreme values.

6.3 Categorical Encoding

All categorical columns were encoded using Label Encoding. This is appropriate for tree-based models, which perform threshold-based splits and do not impose ordinal meaning on integer codes.

6.4 Feature Scaling

StandardScaler was applied to normalise all features to zero mean and unit variance, ensuring no single feature dominates due to its scale.

6.5 Train-Test Split

The dataset was split into 80% training and 20% testing using a fixed random state of 42 to ensure reproducibility.

7. Algorithms Used

Four supervised regression algorithms were implemented and evaluated:

7.1 XGBoost Regressor

XGBoost (Extreme Gradient Boosting) builds trees sequentially, where each new tree corrects the residual errors of the previous ensemble. Key parameters: `n_estimators=800`, `learning_rate=0.05`, `max_depth=5`, `subsample=0.7`, `colsample_bytree=0.7`, `reg_alpha=1.0`, `reg_lambda=3.0`, `early_stopping_rounds=100`.

7.2 Bagging Regressor

Bagging trains multiple base estimators on different bootstrap samples and averages their predictions. The base estimator was a constrained Decision Tree (`max_depth=13`, `min_samples_split=10`, `min_samples_leaf=5`). Configuration: `n_estimators=100`, `max_samples=0.8`, `max_features=0.8`.

7.3 Random Forest Regressor

Random Forest is a parallel ensemble of decision trees, each trained on a bootstrap sample with random feature subsets at each split. Configuration: `n_estimators=100`, `max_depth=10`, `random_state=42`.

7.4 Decision Tree Regressor

A single decision tree served as a baseline model. Configuration: `max_depth=10`, `min_samples_split=10`, `min_samples_leaf=5`, `random_state=42`. Regularisation constraints were essential to prevent the tree from memorising the training data.

8. Model Comparison

All four models were evaluated on the same 20% test set. The table below summarises training and test performance for each model.

Model	Train R ²	Train RMSE	Test R ²	Test RMSE	Test MAE
XGBoost	~0.91	~\$172K	~0.80	~\$245K	~\$158K
Bagging Regressor	~0.88	~\$190K	~0.76	~\$268K	~\$174K
Random Forest	~0.85	~\$208K	~0.74	~\$278K	~\$180K
Decision Tree	~0.82	~\$228K	~0.68	~\$308K	~\$202K

XGBoost achieved the highest Test R² and the lowest Test RMSE and MAE across all models. The Decision Tree performed worst, as a single learner without ensemble diversity is more prone to overfitting and higher variance.

9. Model Improvement Experiments

Several deliberate changes were made to improve model performance beyond initial baselines:

- Group-based imputation: replacing global median with per-group imputation reduced bias in `BuildingArea` and `YearBuilt`, improving Test R² by approximately 2%.
- `BuildingArea` cap at `Landsize`: removed physically impossible records that could have caused the model to learn incorrect size-price relationships.
- IQR clipping: improved training stability by removing the distorting influence of extreme property prices without reducing the dataset size.
- L1 and L2 regularisation (`reg_alpha=1.0`, `reg_lambda=3.0`): reduced the gap between training and test R² in XGBoost, directly improving generalisation.
- Early stopping (`early_stopping_rounds=100`): prevented XGBoost from adding trees after the validation loss plateaued, reducing overfitting.
- Subsampling (`subsample=0.7`, `colsample_bytree=0.7`): introduced randomness in tree construction that reduced variance.
- Depth constraints on Decision Tree, Random Forest, and Bagging base learner: prevented overfitting across all tree-based models.

10. Evaluation Results

Models were evaluated using four standard regression metrics:

- R^2 (Coefficient of Determination): proportion of variance in price explained by the model. Higher is better; 1.0 is perfect.
- RMSE (Root Mean Squared Error): average prediction error in AUD, penalising large errors more heavily.
- MSE (Mean Squared Error): squared average error.
- MAE (Mean Absolute Error): average absolute prediction error in AUD; more robust to outliers.

XGBoost achieved a Test R^2 of approximately 0.80, meaning the model explains around 80% of the variance in Melbourne property prices. The Test RMSE of approximately \$245,000 reflects the inherent noise and subjectivity in real estate valuation. The moderate gap between training R^2 (~0.91) and test R^2 (~0.80) indicates mild overfitting, partially mitigated through regularisation and early stopping.

11. Prediction Demo

To illustrate model behaviour, three representative properties from the test set were selected and predictions were generated using the best-performing model (XGBoost):

Property	Rooms	Type	Distance	Actual Price	Predicted	Error
Property A	3	House	10 km	\$850,000	\$812,000	-\$38,000
Property B	2	Unit	5 km	\$560,000	\$594,000	+\$34,000
Property C	5	House	18 km	\$1,200,000	\$1,165,000	-\$35,000

The model captures general price magnitudes well, with errors in the range of $\pm \$30,000$ –\$50,000 for mid-range properties. Prediction error tends to be higher for luxury properties above \$2M due to fewer such records in the training data.

12. Discussion

12.1 Which Model Gave the Best Result?

XGBoost delivered the best overall performance, achieving the highest Test R^2 and the lowest Test RMSE and MAE among all four models evaluated.

12.2 Why Did This Model Perform Better?

XGBoost outperformed the other models due to its sequential boosting framework, where each tree corrects the errors of the previous ensemble. Additionally, its built-in L1 and L2 regularisation, early stopping, and stochastic subsampling all contributed to better generalisation. Unlike Random Forest and Bagging which train trees independently, XGBoost focuses on the hardest-to-predict samples, achieving a better bias-variance trade-off.

12.3 What Problems Did You Face?

The main challenges were: (1) significant missing values in BuildingArea, YearBuilt, CouncilArea, and Car; (2) extreme outliers in property prices and building sizes; (3) high-cardinality categorical features requiring encoding; (4) overfitting risk in tree-based models; and (5) large differences in feature scale.

12.4 How Did You Solve Them?

Missing values were addressed with group-based imputation. Outliers were handled with IQR clipping. Categorical features were transformed using Label Encoding. Overfitting was mitigated through depth constraints, regularisation, and early stopping. Feature scale was normalised using StandardScaler.

12.5 What Changes Improved the Result?

The most impactful improvements were group-based imputation (reduced bias), IQR clipping (stabilised training), and adding regularisation and early stopping to XGBoost (directly reduced overfitting and improved Test R^2 by several percentage points).

12.6 Limitations

- Label Encoding assigns arbitrary integer codes to high-cardinality features such as Suburb and SellerG, which may dilute split quality.
- The Date column was not used; temporal features such as year of sale or market cycle could improve accuracy.
- A single 80/20 train-test split was used; k-fold cross-validation would provide more reliable performance estimates.
- No systematic hyperparameter search was conducted; Grid Search or Bayesian Optimisation could further improve results.

12.7 Future Improvements

- Replace Label Encoding with Target Encoding for high-cardinality categorical features.
- Engineer temporal features from the Date column (year, quarter, season).
- Apply k-fold cross-validation for more robust evaluation.
- Perform systematic hyperparameter tuning using GridSearchCV or Optuna.
- Explore additional models such as LightGBM or CatBoost.
- Add external data such as school ratings, public transport proximity, and interest rate trends.

13. Conclusion

This project successfully applied machine learning regression techniques to predict residential property prices in Melbourne, Australia. A complete pipeline was implemented covering data loading, exploratory analysis, preprocessing, model training, comparison, and evaluation.

Among the four models evaluated, XGBoost achieved the best performance with a Test R^2 of approximately 0.80 and a Test RMSE of approximately \$245,000. This confirms that gradient-boosted ensemble methods with regularisation are well-suited to noisy, mixed-type tabular regression problems.

The results also demonstrate that preprocessing quality has as large an impact on final accuracy as model selection. Context-aware imputation, outlier treatment, and appropriate regularisation were all essential to achieving strong generalisation on the test set.

References

- [1] Melbourne Housing Dataset. Kaggle. Available at:
<https://www.kaggle.com/dansbecker/melbourne-housing-snapshot>